AUTOMATED TIRE DEFECT DETECTION SYSTEM

A Deep Learning Approach Using MobileNetV2 Transfer Learning

IMAGE CLASSIFICATION PROJECT · INDUSTRIAL AI

EXECUTIVE SUMMARY

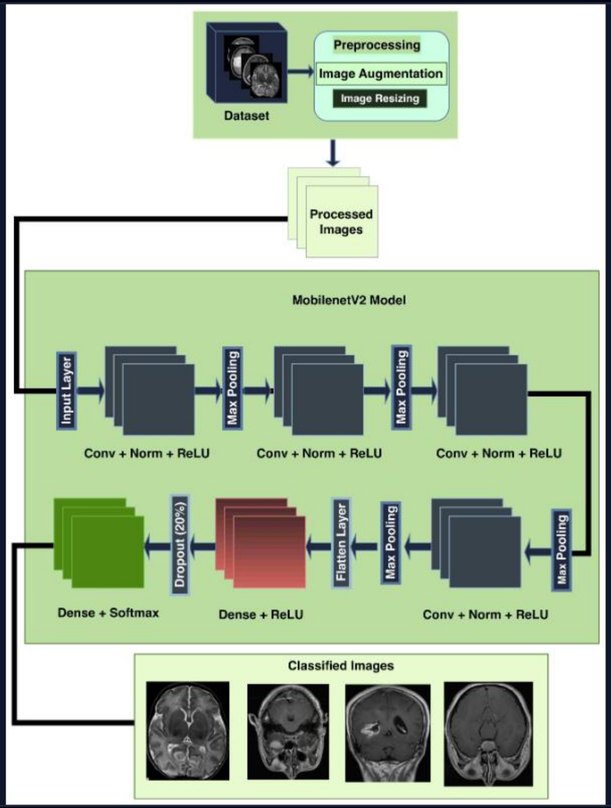
Automating visual inspection in tire manufacturing to identify structural anomalies, reducing manual labor costs and enhancing road safety through high-precision computer vision.

MODEL ARCHITECTURE

TRANSFER LEARNING PIPELINE

By leveraging **MobileNetV2** pre-trained on ImageNet, the model extracts high-level spatial features from tire images effectively.

- **MobileNetV2 Base:** Weights frozen for efficient training.
- **GlobalAvgPool:** Structural dimensionality reduction.
- **Dropout (0.5):** Heavy regularization to prevent overfitting.



DATASET DISTRIBUTION

1,856

Total Images Analyzed

SPLIT & STATISTICS

The dataset is curated for binary classification, ensuring the model sees a balanced mix of "Good" and "Defective" samples.

- **1,484 Training Samples:** Used for model weight adjustment.
- **372 Validation Samples:** Used for real-time monitoring.

PREPROCESSING STRATEGY

ROTATION

Randomly rotating images by 20° to simulate various camera angles in a factory environment.

ZOOM & FLIP

Applying a 0.2 zoom range and horizontal flips to increase variance in the dataset.

RESCALING

Normalizing pixel values from [0, 255] to [0, 1] for faster gradient descent convergence.

ADDRESSING CLASS IMBALANCE

Class weights were calculated to ensure the minority class has a stronger influence on the loss function during training.

Class Label	Condition	Calculated Weight
0	Defective Tire	0.9026
1	Good Tire	1.1208

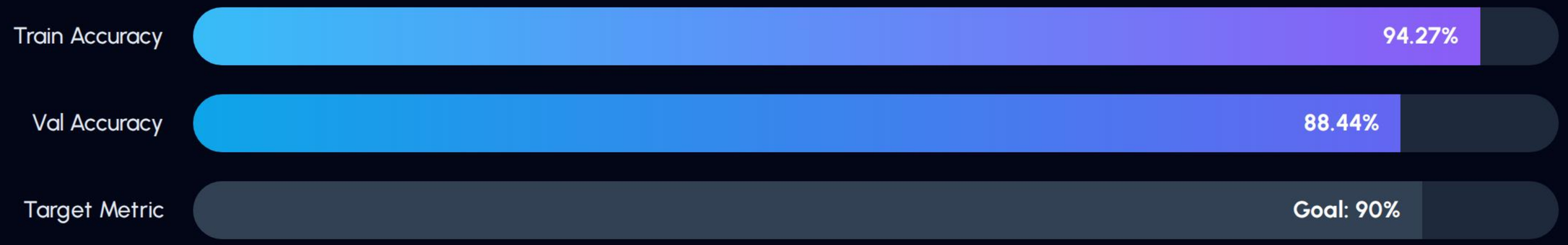
OPTIMIZATION PARAMETERS

- **Adam Optimizer:** Adaptive learning rate optimization.
- **Sigmoid Activation:** Ideal for binary probability output.
- **Loss Function:** Binary Crossentropy used to measure dissimilarity.

BINARY CROSSENTROPY LOSS

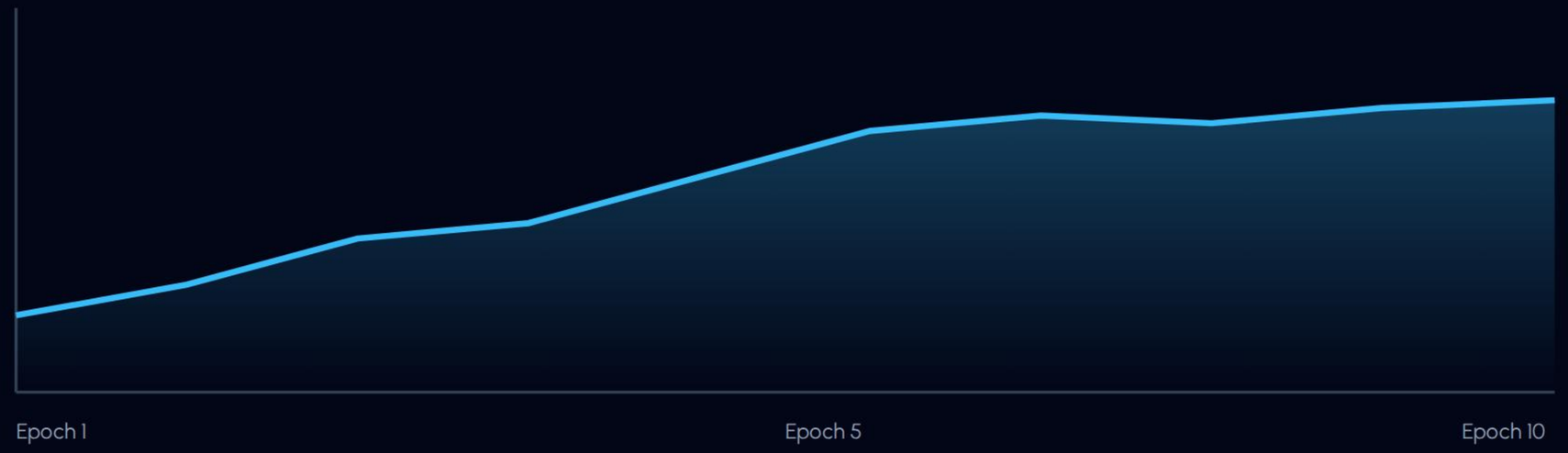
$$L = -\frac{1}{N}\sum_{i=1}^{N} y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i)$$

MODEL PERFORMANCE RESULTS



The model achieved a 94.27% accuracy on training data after 10 epochs, demonstrating strong pattern recognition.

TRAINING PROGRESSION



Model convergence was rapid during the first 6 epochs, followed by stable refinement of weights.

INFERENCE PIPELINE

REAL-TIME DIAGNOSIS

The system processes new tire images (`test.jpg`) through a prediction wrapper that interprets scores as labels.

```
prob = model.predict(img)
label = "Good" if prob > 0.5 else "Defective"
```

Confidence Score Recorded: 96.32%

INDUSTRIAL IMPLICATIONS

PRODUCTION EFFICIENCY

Reducing manual oversight by up to 80%, allowing workers to focus on high-level maintenance rather than repetitive inspection.

OPERATIONAL SAFETY

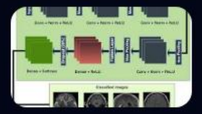
Preventing faulty tires from entering the supply chain, significantly lowering the risk of tire-related road accidents.

QUESTIONS?

Thank you for your attention!

Tire Classification Project · TensorFlow v2.x

IMAGE SOURCES



https://www.mdpi.com/biomedinformatics/biomedinformatics-05-00030/article_deploy/html/images/biomedinformatics-05-00030-g005.png

Source: www.mdpi.com

---

https://irp.cdn-website.com/0fa50680/dms3rep/multi/Screenshot-2023-10-27-1413231.png

Source: www.chapelhilltire.com